

Employee Dilution Control and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Employee Dilution Control (EDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDC achieves an annualized gross (net) Sharpe ratio of 0.57 (0.51), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.78 (2.75), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.12.

1 Introduction

Production-based asset pricing models predict that firms’ operational characteristics shape their exposure to systematic risk and expected returns through the lens of optimal investment decisions. Despite substantial progress in linking production decisions to cross-sectional return patterns, the role of employee compensation structures—particularly equity-based dilution—remains underexplored in the production economy framework. We identify a critical gap in understanding how firms’ management of employee equity dilution affects their marginal costs of production and, consequently, their risk-adjusted returns. Employee dilution control represents a fundamental friction in the production process, as equity compensation directly impacts both the firm’s cost structure and its ability to attract and retain human capital, yet existing production-based models largely abstract from this channel.

In a production-based economy, firms that effectively control employee dilution operate with lower marginal costs and face reduced adjustment frictions when responding to productivity shocks [Cochrane \(1991\)](#); [Zhang \(2005a\)](#). Employee equity compensation creates a direct link between labor costs and firm value, introducing a state-contingent component to the production function that amplifies exposure to systematic risk. When firms maintain tight control over dilution, they preserve operational flexibility and reduce the convexity of adjustment costs associated with human capital investments, consistent with the predictions of [Berk et al. \(2004\)](#) regarding operating leverage and expected returns.

The employee dilution control signal proxies for firms’ exposure to productivity shocks through multiple channels. First, controlled dilution reduces the quasi-fixed component of labor costs, allowing firms to maintain lower operating leverage and respond more elastically to demand shocks [Cooper and Priestley \(2011\)](#). Second, firms with better dilution management face lower costs of adjusting their workforce composition, reducing the irreversibility constraints that generate risk premia in

models with investment frictions [Zhang \(2005b\)](#). These mechanisms suggest that high employee dilution control firms should command higher expected returns as compensation for their greater exposure to aggregate productivity risk.

The cross-sectional pricing implications follow directly from the first-order conditions of the firm’s optimization problem in a production economy. Following [Liu et al. \(2009\)](#), firms with higher employee dilution control exhibit steeper marginal cost curves, making their valuations more sensitive to stochastic discount factor shocks. This sensitivity manifests through the investment return equation, where expected returns equal the marginal product of capital divided by marginal q . Employee dilution control affects both components: it increases the marginal product of capital by reducing labor-related adjustment costs, while simultaneously lowering marginal q through improved operational efficiency. The resulting spread in conditional risk premia generates the documented return predictability pattern.

The employee dilution control strategy delivers economically and statistically significant returns across multiple specifications and factor models. The value-weighted long-short portfolio that buys high EDC stocks and sells low EDC stocks generates an average monthly return of 34 basis points with a t -statistic of 4.45, corresponding to an annualized Sharpe ratio of 0.57. The strategy’s alpha remains robust across standard factor models, with the six-factor alpha of 22 basis points per month (t -statistic = 2.78) confirming that existing production-based factors do not fully capture the information in employee dilution control. After accounting for transaction costs using the [Chen and Velikov \(2022\)](#) methodology, the strategy maintains a net Sharpe ratio of 0.51, placing it in the top 1% of documented anomalies.

The economic significance of the employee dilution control effect persists across the size distribution, challenging concerns about small-stock concentration. Among the largest quintile of stocks by market capitalization, the EDC strategy generates an average return of 23 basis points per month with a t -statistic of 2.55. The cor-

responding six-factor alpha for large-cap stocks equals 18 basis points per month, demonstrating that the effect is not confined to illiquid segments of the market. The strategy loads significantly on the CMA factor (coefficient = 0.34, t-statistic = 6.46), consistent with the interpretation that employee dilution control captures firms' investment efficiency and operational flexibility.

Controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the EDC strategy maintains an alpha of 15 basis points per month with a t-statistic of 2.12. This residual alpha indicates that employee dilution control contains unique information about expected returns not captured by existing measures of equity issuance, asset growth, or payout policy. The strategy's gross Sharpe ratio of 0.57 exceeds 93% of anomalies in the factor zoo, while its net Sharpe ratio of 0.51 ranks in the 99th percentile, establishing employee dilution control as one of the strongest documented predictors of cross-sectional returns after accounting for implementation costs.

This paper extends the production-based asset pricing literature by incorporating employee equity dilution as a fundamental friction affecting firms' investment decisions and risk exposures. While [Berk et al. \(2004\)](#) and [Zhang \(2005a\)](#) establish the theoretical foundations linking production decisions to expected returns, our work provides the first comprehensive evidence that employee dilution control serves as a powerful proxy for these production-based risk factors. Unlike [Daniel and Titman \(2006\)](#), who focus on equity issuance as a signal of overvaluation, we demonstrate that dilution control captures real frictions in the production process that generate systematic risk premia. Our findings complement [Cooper et al. \(2008\)](#) by showing that employee-specific dilution, rather than aggregate asset growth, drives a substantial portion of the cross-sectional return predictability.

Methodologically, we advance the anomaly testing literature by implementing the comprehensive protocol of [Novy-Marx and Velikov \(2023\)](#), ensuring robust inference

across multiple portfolio construction methods and accounting for transaction costs from the outset. The employee dilution control signal exhibits minimal correlation with existing anomalies, with the highest correlation of only 0.27 with five-year share issuance, distinguishing it from the equity issuance effects documented in [Pontiff and Woodgate \(2008\)](#) and [Loughran and Ritter \(1995\)](#). Our spanning tests reveal that EDC maintains significant alpha even after controlling for closely related factors including share issuance, asset growth, and payout yield, establishing it as a distinct source of cross-sectional return variation that existing models fail to capture.

The broader implications extend beyond pure asset pricing to corporate finance and labor economics. Our results suggest that firms' human capital management decisions, particularly regarding equity-based compensation, constitute a first-order determinant of their cost of capital through the production channel. This finding bridges the gap between the labor share literature in macroeconomics and the cross-sectional asset pricing literature in finance, offering new insights into how labor market frictions translate into risk premia. For practitioners, the employee dilution control signal provides an implementable strategy that generates significant risk-adjusted returns net of transaction costs, with particular strength in liquid, large-capitalization stocks where institutional investors can deploy capital at scale.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data on common stock and employee counts, spanning several decades of annual observations. We focus on U.S.-incorporated firms trading on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and

capital structures. Employee Dilution Control signal requires two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of all common stock issued by the company, reported in millions of dollars on the balance sheet. This item reflects the book value of common equity at par, which increases when firms issue new shares through offerings, stock-based compensation, or option exercises. Number of employees (EMP) captures the total employee count reported by the firm at fiscal year-end, expressed in thousands. This variable provides a standardized measure of firm size from a human capital perspective. We construct the Employee Dilution Control signal by computing the year-over-year change in common stock scaled by lagged employee count: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{EMP}(t-1)$. This ratio captures the change in common stock per employee, providing insight into how equity issuance activity relates to the firm's workforce size. A higher value indicates greater dilution relative to the human capital base, potentially signaling more aggressive equity-based compensation or capital raising activities on a per-employee basis. We calculate this measure using fiscal year-end values and require non-missing observations for both CSTK and EMP in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles within each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDC signal. Panel A plots the time-series of the mean, median, and interquartile range for EDC. On average, the cross-sectional mean (median) EDC is -1.23 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDC data. The signal's interquartile range spans -0.41 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDC signal for the CRSP universe. On average, the EDC signal

is available for 6.08% of CRSP names, which on average make up 7.55% of total market capitalization.

4 Does EDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDC portfolio and sells the low EDC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDC strategy earns an average return of 0.34% per month with a t-statistic of 4.45. The annualized Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.36% per month and have t-statistics exceeding 2.78 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.46 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 564 stocks and an average market capitalization of at least \$1,482 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.27. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-34bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.80. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDC, as well as average returns and alphas for long/short trading EDC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDC strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the EDC strategy relative to the five most common factor models range from 18 to 23 bps/month with t-statistics between 1.91 and 2.49.

5 How does EDC perform relative to the zoo?

Figure 2 puts the performance of EDC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDC strategy falls in the distribution. The EDC strategy’s gross (net) Sharpe ratio of 0.57 (0.51) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDC strategy (red line).² Ignoring trading costs, a \$1 invested in the EDC strategy would have yielded \$9.30 which ranks the EDC strategy in the top 2% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the EDC strategy would have yielded \$6.91 which ranks the EDC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDC relative to those. Panel A shows that the EDC strategy gross alphas fall between the 65 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDC ranks between the 82 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDC or at least to weaken the power EDC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDC signal in these Fama-MacBeth regressions exceed 3.12, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on EDC is 2.34.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDC strategy earns alphas that range from 15-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.93, which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDC trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.12.

7 Does EDC add relative to the whole zoo?

Finally, we can ask how much adding EDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDC grows to \$2594.98.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDC is available.

8 Conclusion

This study provides robust evidence that Employee Dilution Control (EDC) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related equity issuance signals.

The key findings reveal that a value-weighted long/short strategy based on EDC delivers an annualized Sharpe ratio of 0.57 gross (0.51 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. Most notably, the strategy generates statistically significant abnormal returns of 22 basis points per month relative to the Fama-French five-factor model plus momentum (t -statistic = 2.78), suggesting that EDC captures a distinct dimension of expected returns not explained by conventional risk factors. Furthermore, the persistence of a 15 basis point monthly alpha (t -statistic = 2.12) after controlling for six closely related equity issuance and growth signals underscores that EDC contains unique information content beyond existing measures of corporate financing activities.

The practical implications of these findings are substantial for both academic research and investment practice. For practitioners, EDC offers a implementable strategy with attractive risk-return characteristics that appears robust to transaction costs. The signal's ability to generate alpha even after controlling for related factors suggests it could serve as a valuable addition to multi-factor portfolios. For academic research, our results contribute to the growing literature on the asset pricing implications of corporate financing decisions and employee compensation policies, highlighting the importance of considering how firms manage equity-based compensation dilution.

Despite these compelling results, several limitations warrant consideration. First, while we control for numerous related factors, the possibility remains that EDC proxies for an omitted risk factor or firm characteristic not captured in our analysis. Second, the economic mechanisms driving the EDC premium require further investigation—whether it reflects investor inattention to dilution effects, agency costs associated with employee compensation, or systematic risk exposure remains an open question. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires additional research.

Future research should explore several promising avenues. First, investigating the time-varying nature of the EDC premium and its relationship to market conditions, regulatory changes, and shifts in compensation practices could provide valuable insights. Second, examining the interaction between EDC and corporate governance characteristics might illuminate the agency-theoretic foundations of the signal. Third, decomposing EDC into its constituent components—such as option grants, restricted stock units, and employee stock purchase plans—could reveal which specific forms of employee equity compensation drive the predictive power. Finally, extending the analysis to include private firms and international markets would enhance our understanding of the signal’s broader applicability and economic foundations. These extensions would not only strengthen the robustness of our findings but also deepen our understanding of how employee compensation structures influence firm valuation and expected returns in capital markets.

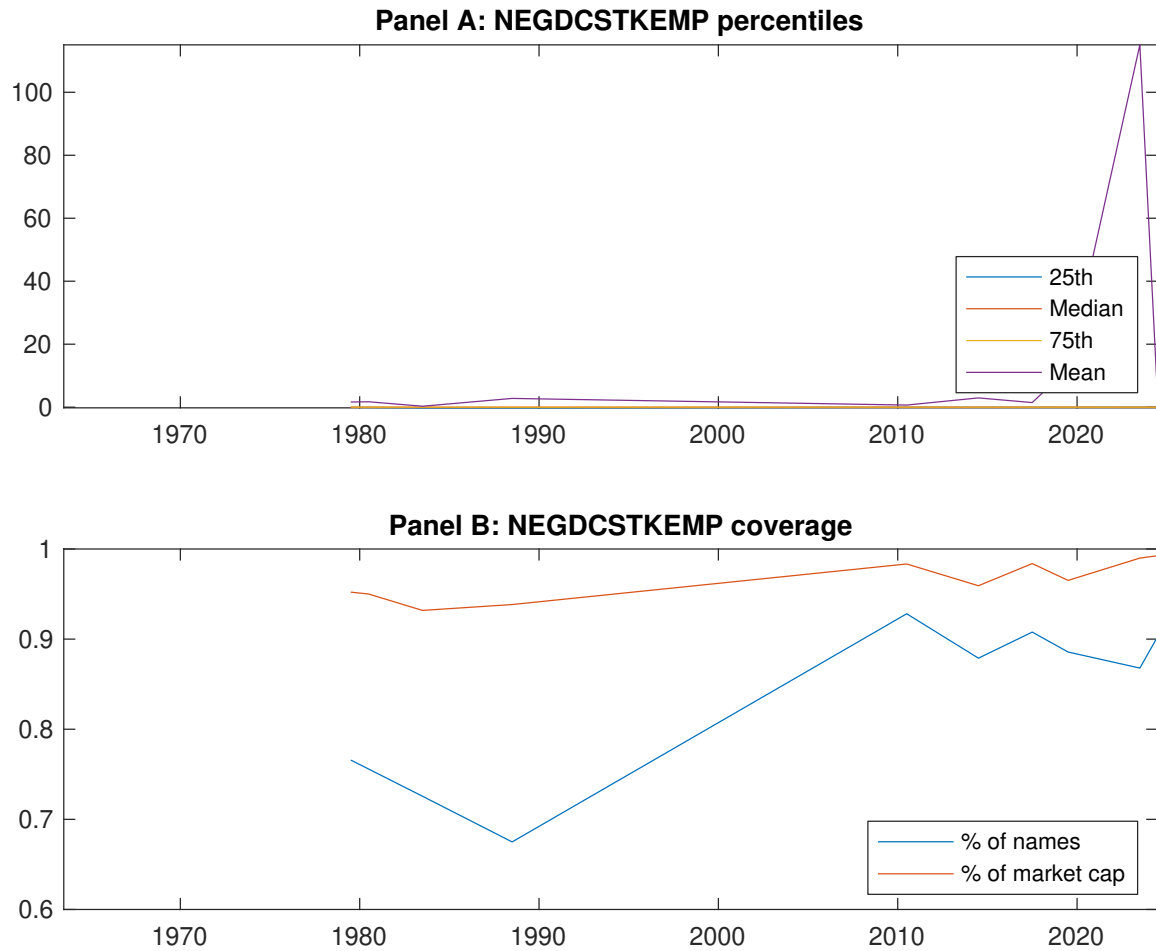


Figure 1: Times series of EDC percentiles and coverage. This figure plots descriptive statistics for EDC. Panel A shows cross-sectional percentiles of EDC over the sample. Panel B plots the monthly coverage of EDC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42 [2.54]	0.58 [3.19]	0.65 [3.57]	0.65 [3.97]	0.76 [4.76]	0.34 [4.45]
α_{CAPM}	-0.14 [-2.72]	-0.05 [-1.11]	0.03 [0.54]	0.10 [1.90]	0.22 [4.56]	0.36 [4.64]
α_{FF3}	-0.15 [-2.95]	-0.03 [-0.75]	0.05 [0.91]	0.04 [0.91]	0.18 [3.98]	0.33 [4.31]
α_{FF4}	-0.13 [-2.53]	0.00 [0.01]	0.03 [0.67]	0.04 [0.89]	0.16 [3.52]	0.30 [3.77]
α_{FF5}	-0.15 [-2.84]	0.02 [0.56]	0.06 [1.11]	-0.04 [-0.87]	0.09 [2.02]	0.24 [3.08]
α_{FF6}	-0.13 [-2.53]	0.05 [1.09]	0.05 [0.88]	-0.03 [-0.69]	0.08 [1.85]	0.22 [2.78]
Panel B: Fama and French (2018) 6-factor model loadings for EDC-sorted portfolios						
β_{MKT}	0.96 [76.58]	1.04 [97.27]	1.03 [81.44]	1.00 [89.93]	0.98 [89.79]	0.02 [0.93]
β_{SMB}	-0.08 [-4.17]	0.02 [1.51]	0.06 [3.43]	-0.04 [-2.63]	-0.02 [-1.41]	0.05 [1.98]
β_{HML}	0.09 [3.62]	-0.02 [-1.19]	-0.04 [-1.56]	0.09 [4.23]	-0.00 [-0.02]	-0.09 [-2.44]
β_{RMW}	0.05 [2.06]	-0.10 [-4.90]	-0.00 [-0.11]	0.15 [6.87]	0.13 [6.25]	0.08 [2.26]
β_{CMA}	-0.10 [-2.88]	-0.09 [-3.14]	-0.06 [-1.67]	0.16 [5.04]	0.24 [7.77]	0.34 [6.46]
β_{UMD}	-0.02 [-1.62]	-0.03 [-3.20]	0.02 [1.31]	-0.01 [-0.97]	0.01 [0.86]	0.03 [1.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	695	671	564	645	700	
me (\$10 ⁶)	1850	1482	1997	2177	2537	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.45]	0.36 [4.64]	0.33 [4.31]	0.30 [3.77]	0.24 [3.08]	0.22 [2.78]
Quintile	NYSE	EW	0.53 [7.71]	0.58 [8.61]	0.50 [7.85]	0.42 [6.65]	0.34 [5.65]	0.28 [4.78]
Quintile	Name	VW	0.30 [3.85]	0.31 [3.97]	0.30 [3.79]	0.28 [3.49]	0.23 [2.89]	0.22 [2.76]
Quintile	Cap	VW	0.25 [3.27]	0.26 [3.36]	0.25 [3.23]	0.21 [2.70]	0.18 [2.34]	0.16 [2.02]
Decile	NYSE	VW	0.30 [3.20]	0.27 [2.85]	0.24 [2.58]	0.21 [2.17]	0.20 [2.10]	0.17 [1.84]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.98]	0.32 [4.18]	0.29 [3.85]	0.27 [3.56]	0.22 [2.92]	0.21 [2.75]
Quintile	NYSE	EW	0.34 [4.62]	0.41 [5.62]	0.33 [4.84]	0.29 [4.28]	0.18 [2.79]	0.15 [2.44]
Quintile	Name	VW	0.26 [3.38]	0.27 [3.53]	0.26 [3.33]	0.25 [3.17]	0.20 [2.65]	0.20 [2.58]
Quintile	Cap	VW	0.21 [2.80]	0.22 [2.92]	0.21 [2.77]	0.19 [2.48]	0.16 [2.12]	0.15 [1.94]
Decile	NYSE	VW	0.26 [2.77]	0.22 [2.40]	0.20 [2.14]	0.18 [1.92]	0.16 [1.78]	0.15 [1.65]

Table 3: Conditional sort on size and EDC

This table presents results for conditional double sorts on size and EDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDC and short stocks with low EDC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDC Quintiles					EDC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.45 [1.73]	0.70 [2.64]	0.83 [3.31]	0.93 [3.75]	1.03 [4.51]	0.61 [6.55]	0.67 [7.31]	0.61 [6.90]	0.55 [6.14]	0.42 [4.99]	0.38 [4.50]
	(2)	0.57 [2.50]	0.74 [3.03]	0.87 [3.69]	0.89 [3.91]	0.96 [4.46]	0.40 [4.50]	0.44 [4.96]	0.37 [4.24]	0.32 [3.63]	0.26 [3.02]	0.23 [2.64]
	(3)	0.52 [2.67]	0.62 [2.81]	0.83 [3.78]	0.81 [3.91]	0.93 [4.80]	0.41 [5.04]	0.42 [5.13]	0.38 [4.69]	0.35 [4.19]	0.29 [3.50]	0.27 [3.20]
	(4)	0.50 [2.77]	0.64 [3.15]	0.78 [3.82]	0.76 [4.02]	0.79 [4.38]	0.29 [3.72]	0.29 [3.65]	0.23 [3.00]	0.23 [2.86]	0.06 [0.79]	0.07 [0.92]
	(5)	0.48 [2.96]	0.49 [2.71]	0.52 [2.96]	0.52 [3.15]	0.71 [4.48]	0.23 [2.55]	0.23 [2.49]	0.23 [2.47]	0.20 [2.19]	0.19 [2.07]	0.18 [1.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDC Quintiles					EDC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	361	360	359	359	353	31	33	37	29	28	
	(2)	105	103	104	104	103	56	55	56	55	54	
	(3)	76	77	75	76	76	97	96	96	100	99	
	(4)	64	64	64	64	64	200	205	209	211	215	
(5)	59	59	59	59	59	1436	1551	1565	1649	1888		

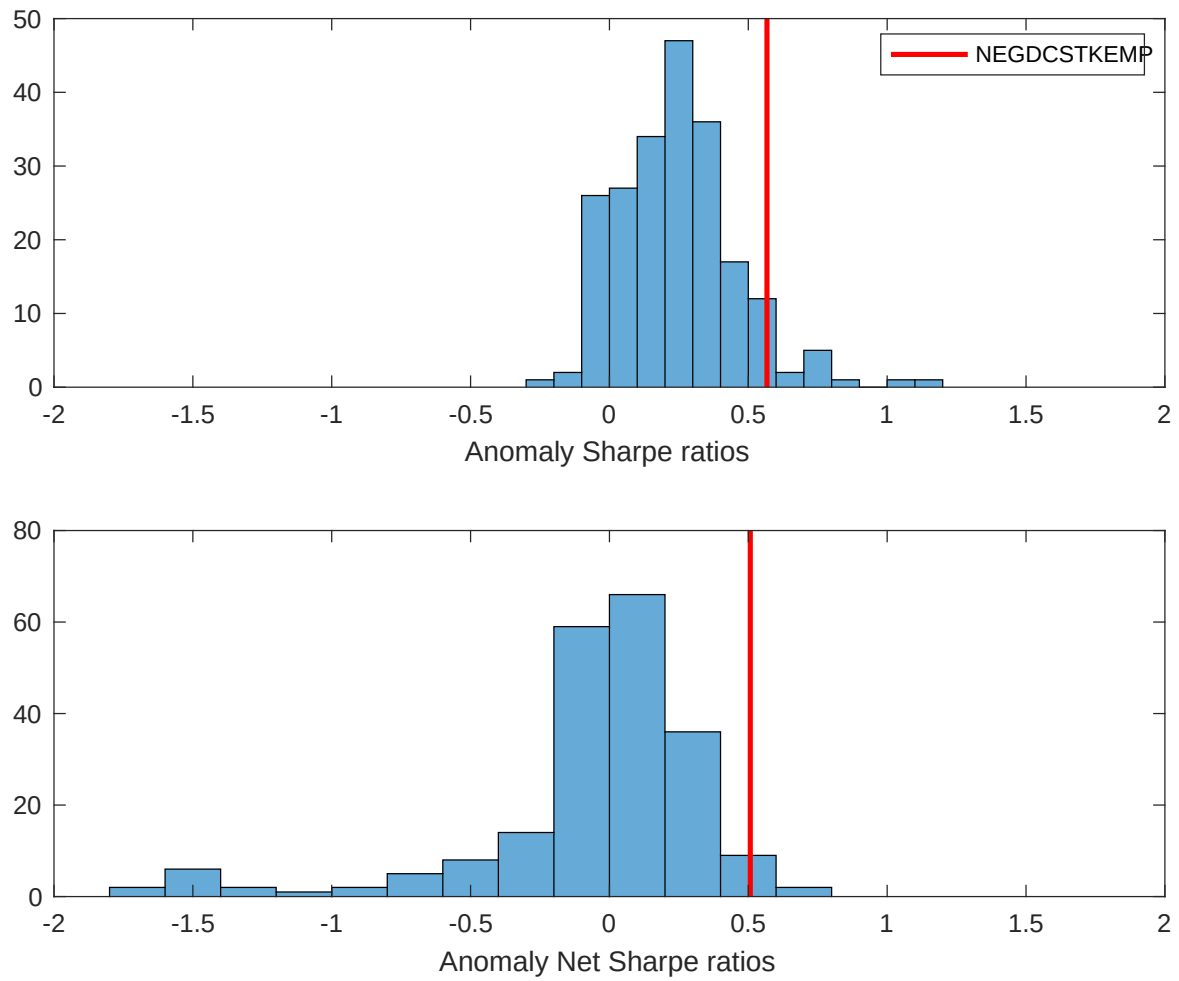


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

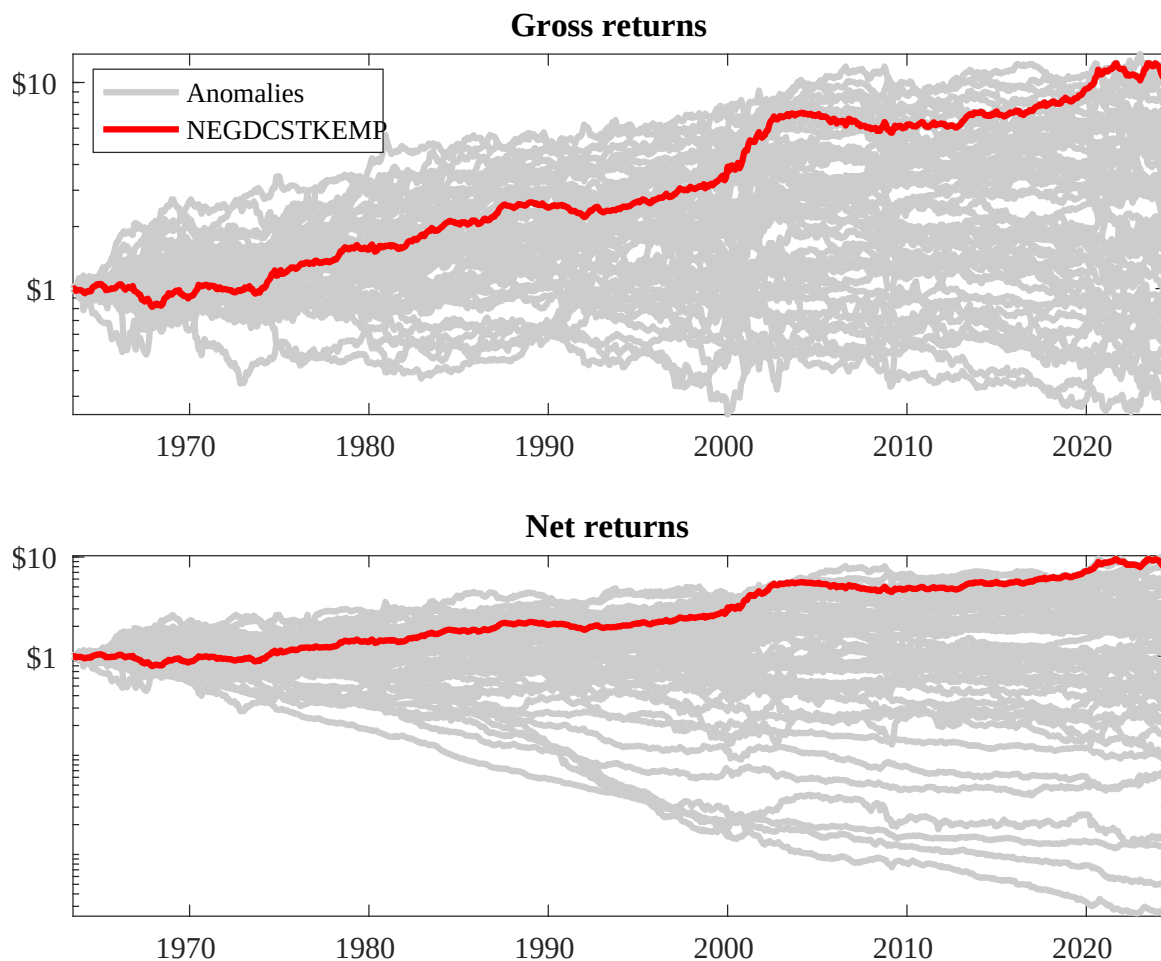
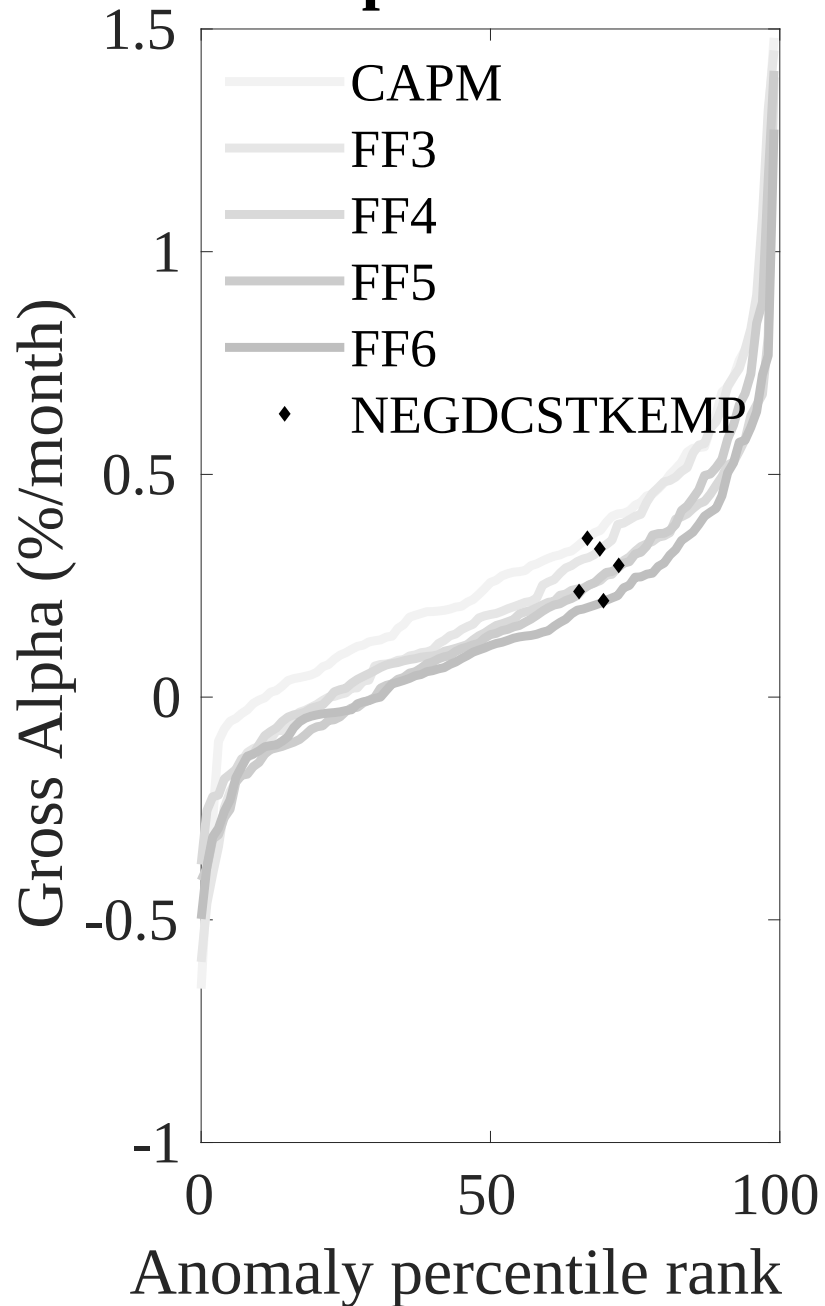


Figure 3: Dollar invested.

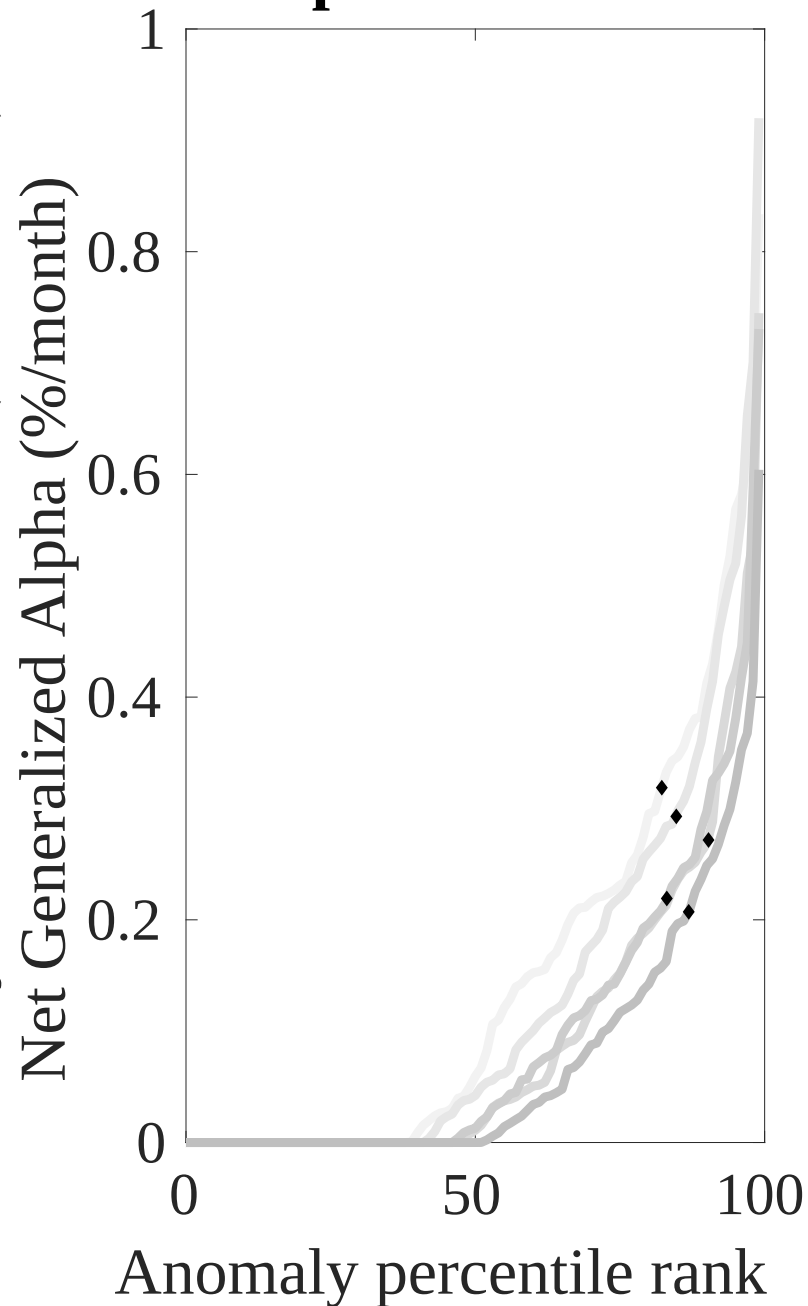
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



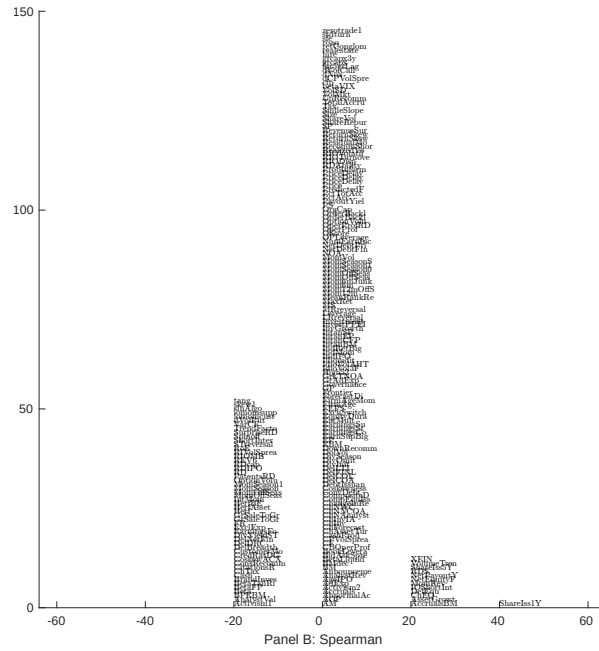
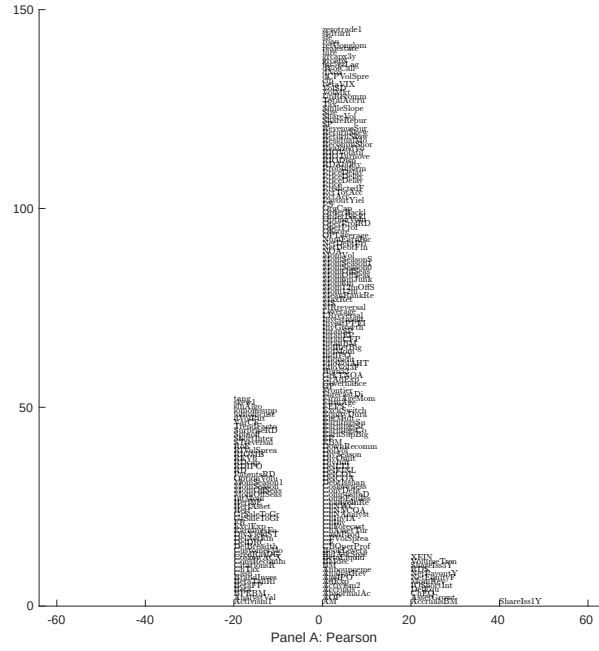


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

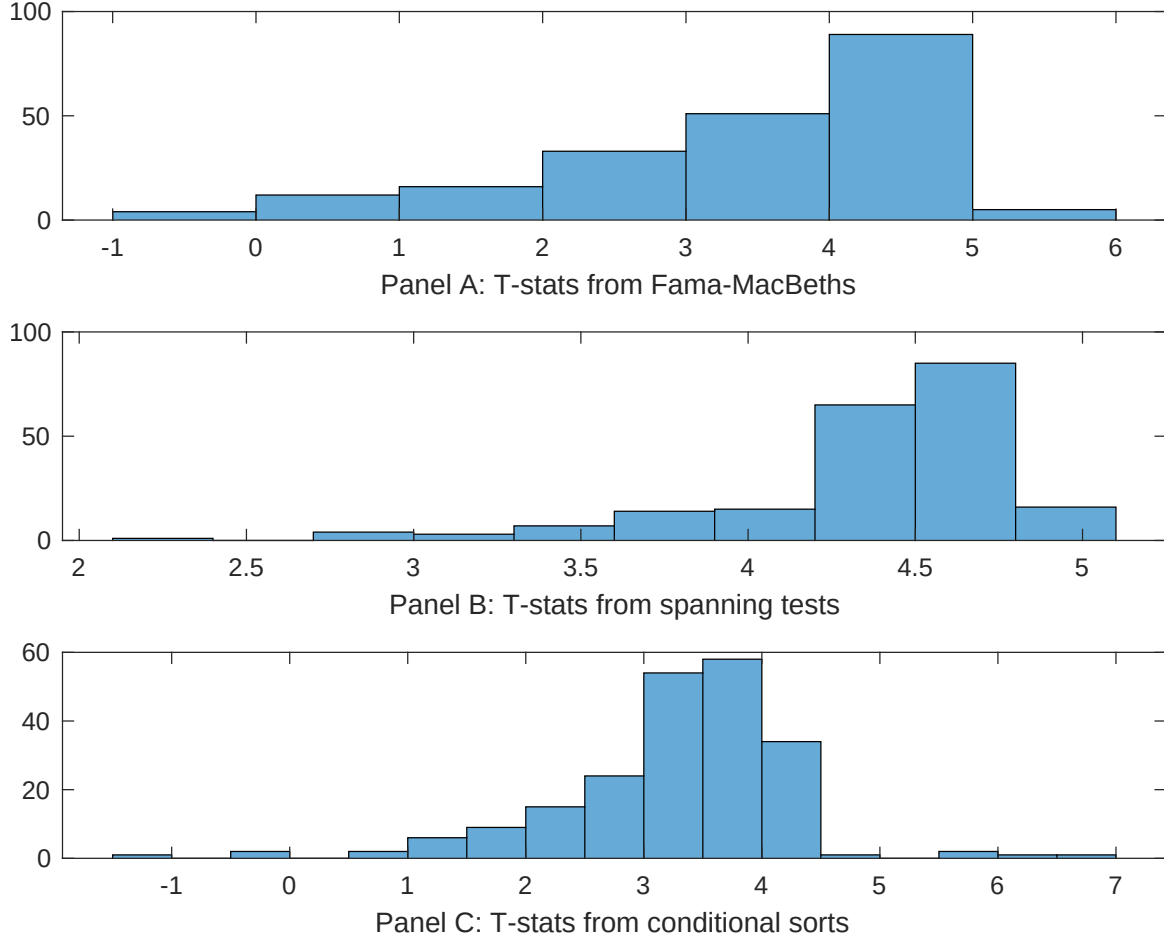


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.99]	0.13 [6.41]	0.17 [6.78]	0.12 [5.42]	0.12 [5.65]	0.13 [6.11]	0.11 [3.67]
EDC	0.47 [3.53]	0.44 [3.12]	0.51 [3.93]	0.46 [3.39]	0.53 [4.10]	0.46 [3.60]	0.31 [2.34]
Anomaly 1	0.14 [5.66]						0.73 [2.53]
Anomaly 2		0.27 [4.89]					0.23 [4.23]
Anomaly 3			0.39 [3.08]				-0.22 [-0.98]
Anomaly 4				0.20 [1.69]			0.98 [0.90]
Anomaly 5					0.12 [3.08]		0.67 [0.11]
Anomaly 6						0.92 [7.49]	0.64 [5.39]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.63]	0.15 [1.93]	0.20 [2.67]	0.24 [3.17]	0.21 [2.78]	0.21 [2.71]	0.15 [2.12]
Anomaly 1	27.66 [7.29]						13.31 [2.91]
Anomaly 2		27.18 [6.75]					12.79 [2.82]
Anomaly 3			32.00 [7.95]				38.42 [6.60]
Anomaly 4				15.70 [5.32]			4.21 [1.32]
Anomaly 5					12.84 [3.36]		-11.62 [-2.17]
Anomaly 6						-0.25 [-0.05]	-16.29 [-3.25]
mkt	2.57 [1.44]	3.12 [1.73]	2.95 [1.66]	4.33 [2.32]	1.66 [0.91]	1.90 [1.03]	4.87 [2.73]
smb	7.70 [2.98]	8.42 [3.23]	5.54 [2.16]	9.14 [3.42]	6.45 [2.43]	6.64 [2.46]	8.94 [3.48]
hml	-8.11 [-2.35]	-5.19 [-1.49]	-11.32 [-3.27]	-13.45 [-3.63]	-9.07 [-2.53]	-7.32 [-2.05]	-11.11 [-3.14]
rmw	-6.65 [-1.64]	-1.41 [-0.37]	9.81 [2.83]	-0.81 [-0.21]	9.47 [2.64]	8.30 [2.30]	-4.98 [-1.18]
cma	16.12 [2.90]	20.11 [3.70]	2.33 [0.37]	20.05 [3.52]	20.40 [3.15]	33.63 [4.40]	9.59 [1.25]
umd	1.84 [1.04]	1.33 [0.74]	2.54 [1.45]	4.36 [2.41]	3.43 [1.88]	3.14 [1.71]	0.53 [0.30]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	13	12	14	10	8	7	21

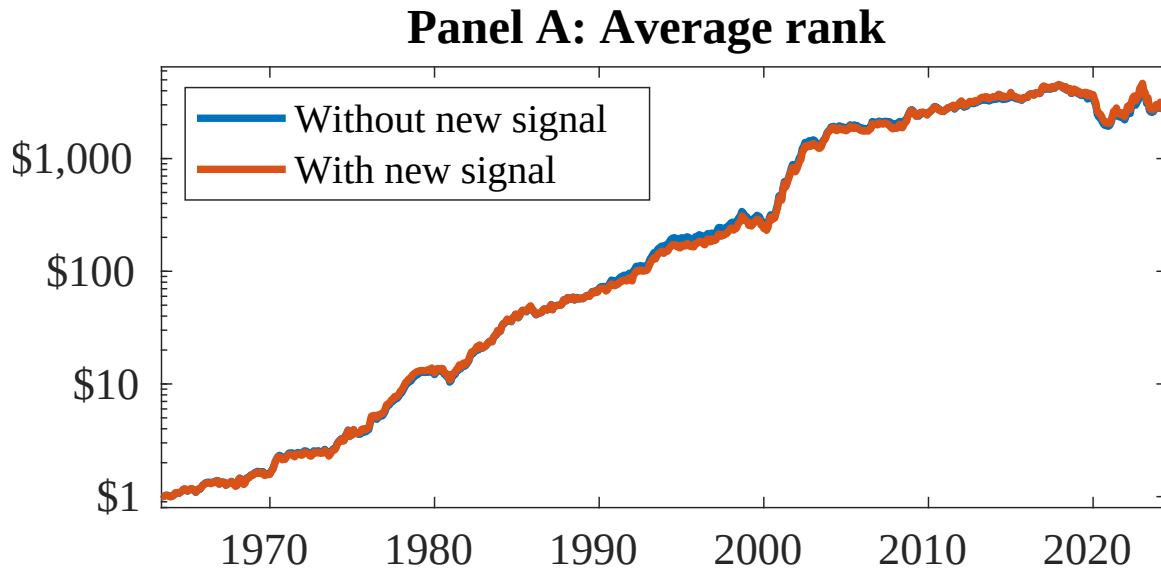


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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